

Homework — 1.4.1 Data Types — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [3] 1 mark per fully correct row:

Denary	8-bit binary	Hexadecimal
156	(a) 10011100	(b) 9C
(c) 202	11001010	(d) CA
111	(e) 01101111	6F

Row 1: $156 = 128 + 16 + 8 + 4 \rightarrow 10011100$; nibbles $1001 = 9$, $1100 = C \rightarrow 9C$. Row 2: $11001010 = 128 + 64 + 8 + 2 = 202$; nibbles $1100 = C$, $1010 = A \rightarrow CA$. Row 3: $6F = (6 \times 16) + 15 = 111$; $6 = 0110$, $F = 1111 \rightarrow 01101111$ (check: $64 + 32 + 8 + 4 + 2 + 1 = 111$).

2. [4] $52 - 37 = 52 + (-37)$. $+52 = 00110100$ ($32 + 16 + 4 = 52$). $\checkmark +37 = 00100101$ ($32 + 4 + 1 = 37$). $\checkmark -37$: invert $00100101 \rightarrow 11011010$, then $+1 \rightarrow 11011011$. Add:

```
00110100    (52)
+ 11011011   (-37)
-----
1 00001111
```

The carry out of the most significant bit is **discarded** (the ninth bit is dropped), leaving $00001111 = 8 + 4 + 2 + 1 = 15$. $\checkmark (52 - 37 = 15)$ Mark: $+52$ and $+37$ correct in binary (1); correct two's complement of $37 = 11011011$ (1); correct binary addition giving $1\ 00001111$ (1); states the final carry is discarded, giving $00001111 = 15$ (1).

3. [2] (2 correct rows per mark.)

- Row 1: **True** — 01001100 (76) $\rightarrow 00100110 \rightarrow 00010011$ (19), and $76 \div 4 = 19$; the sign bit 0 is copied in from the left.
- Row 2: **False** — ANDing with 00001111 **clears the four leftmost bits to 0** (and keeps the four rightmost bits unchanged).
- Row 3: **True**.
- Row 4: **False** — XORing a bit with 1 **flips (inverts)** the bit; it is XORing with 0 that leaves it unchanged.

4. [3] Character set = **C**; ASCII = **F**; Unicode = **D**; Overflow = **A**; Mantissa = **B**; Exponent = **E**. (2 correct matches per mark.)

5. [2] Integers are stored exactly as whole binary numbers, whereas real numbers are stored in floating point as a mantissa and exponent, which can only *approximate* many values (1). So storing the score as a real risks rounding/representation errors (e.g. comparisons failing / the stored value not being exactly the

score), and it typically wastes storage and is slower to process — an integer stores every whole-number score exactly (1). *Max 2: one representational point plus one consequence.*

Part B (6 marks)

6. [1] The MAR holds the **address** of the memory location about to be read from or written to — during fetch, the address of the next instruction copied from the PC (1).

7. [3] Virtual memory uses an area of **secondary storage** (a page/swap file) as if it were an extension of RAM (1). When RAM is full, pages of a program not currently in use are moved out to disk, and moved back into RAM when they are needed (1). This allows a program larger than physical RAM to run, although accessing swapped-out pages is much slower than RAM (excessive swapping = disk thrashing) (1).

8. [2] Difference (1): lossless compression allows the original file to be reconstructed *exactly*, whereas lossy compression permanently discards some data, so the original cannot be perfectly recovered. Uses (1): lossless for text/source code/program files (or PNG/ZIP), where every bit matters; lossy for photographs/audio/video (e.g. JPEG, MP3), where a small quality loss is acceptable in exchange for a much smaller file.

Total: 14 + 6 = 20 marks